

Physical Chemistry (I)

Lecture 02

Basic Concepts

Perfect Gas



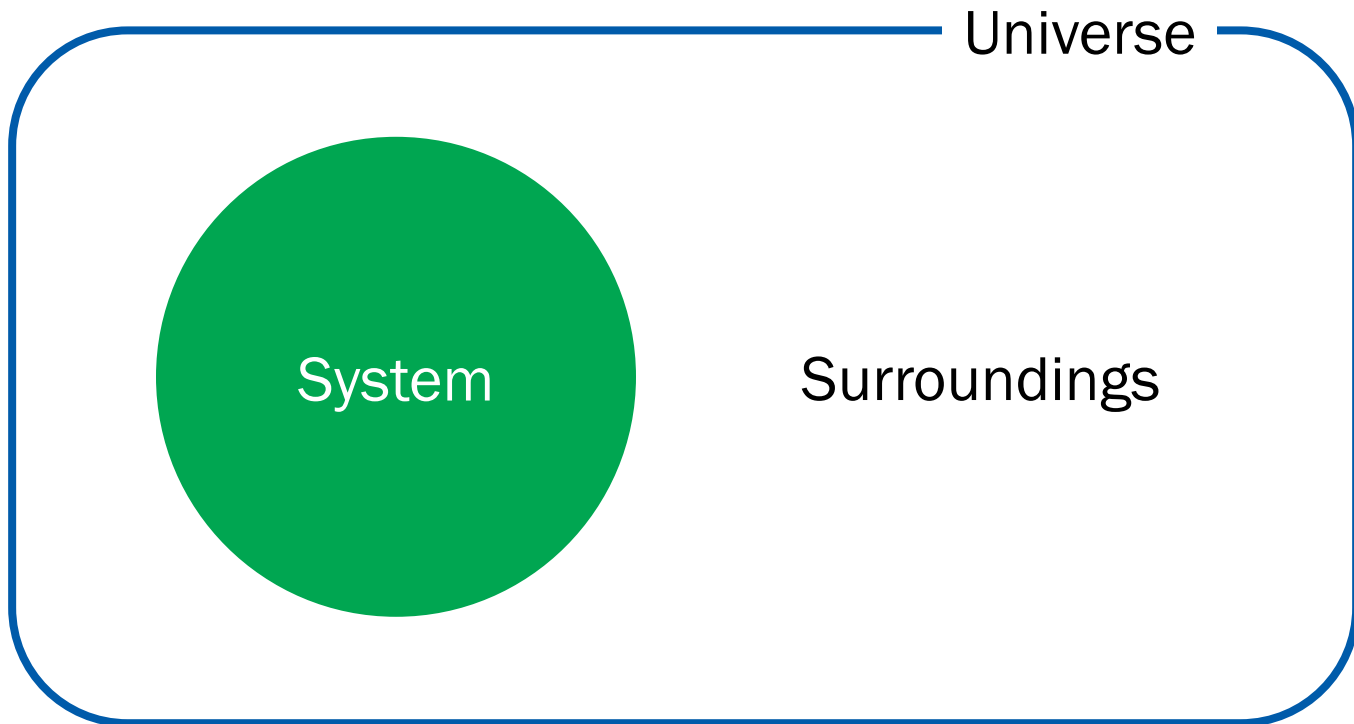
“Chapters” in This Course

1. Basic concepts (1, 2)
2. Gases (2)
3. The first law (1)
4. Entropy (3)
5. Helmholtz and Gibbs energies (3)
6. Statistical mechanics (X)
7. Phase transitions (4)
8. Simple mixtures (5)
9. Chemical equilibria (6)
10. Electrochemistry (6)

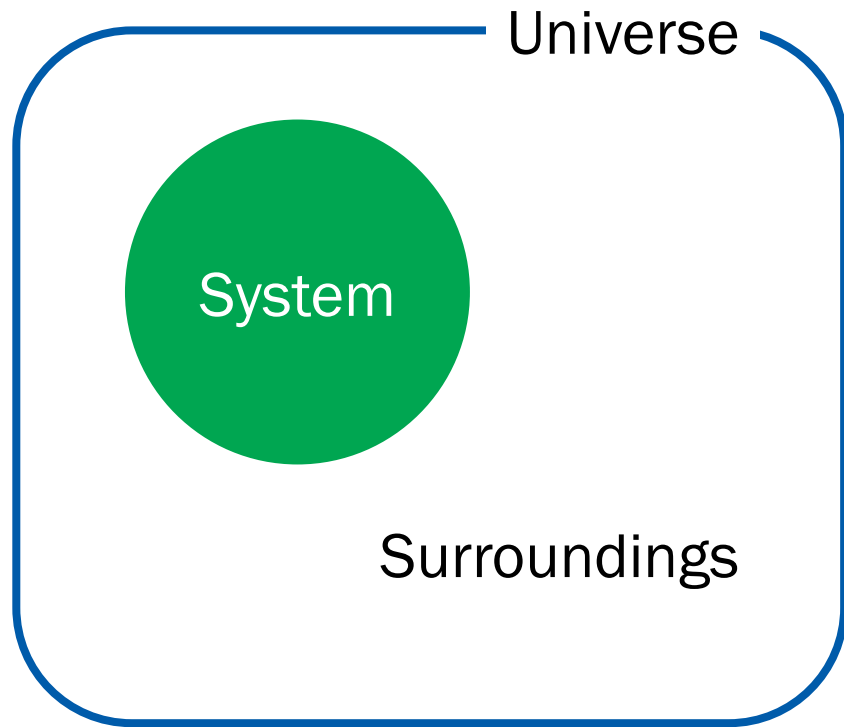


1. Basic Concepts

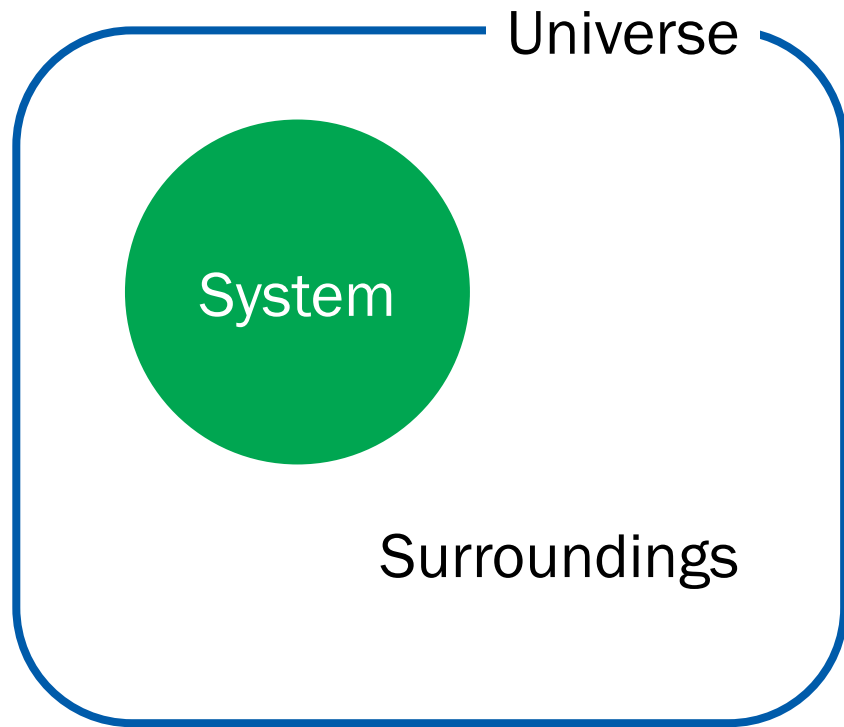
System and Surroundings



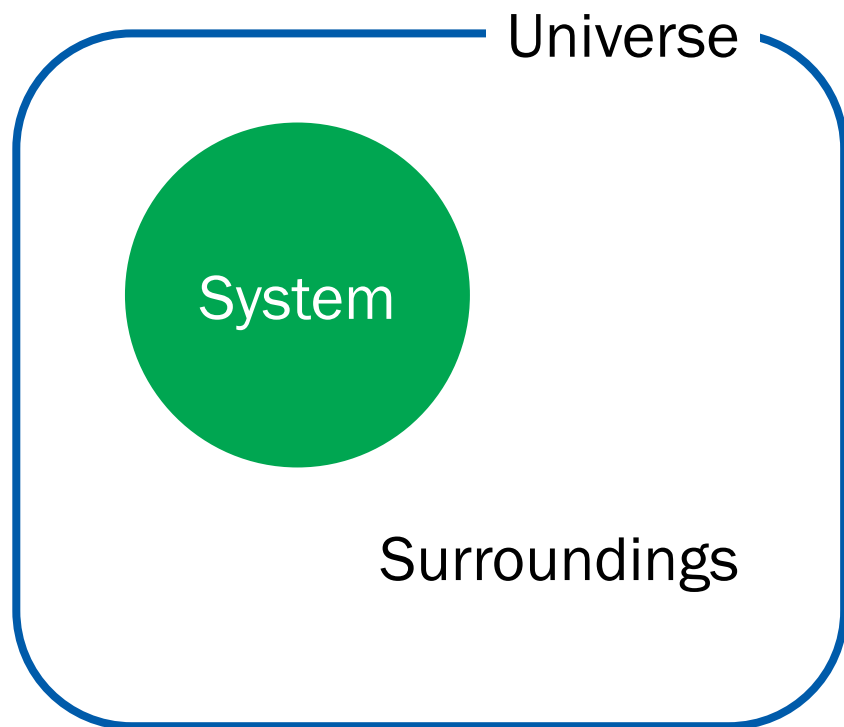
Example of a System



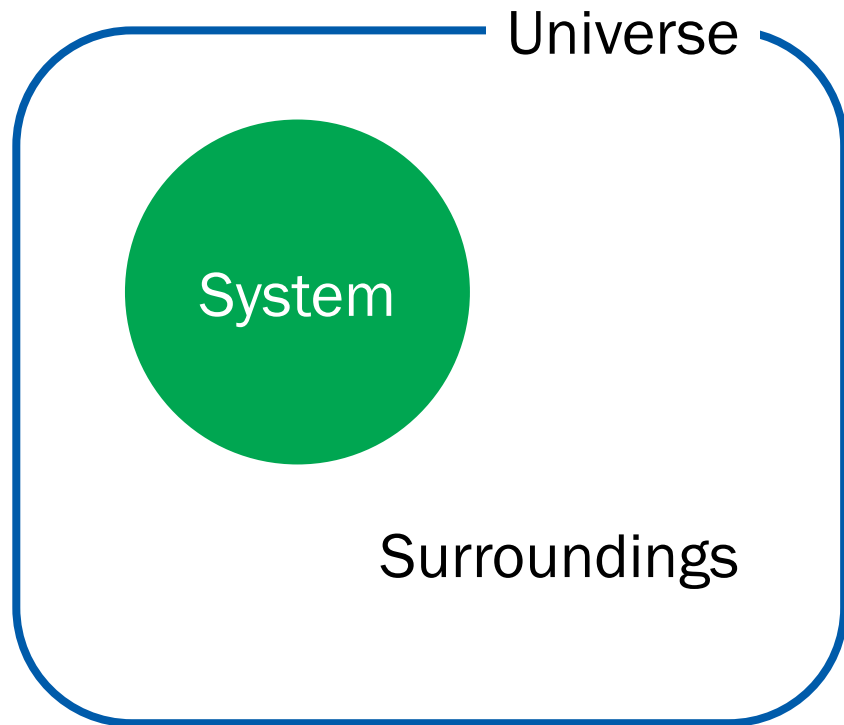
Example of a System



Another Example of a System

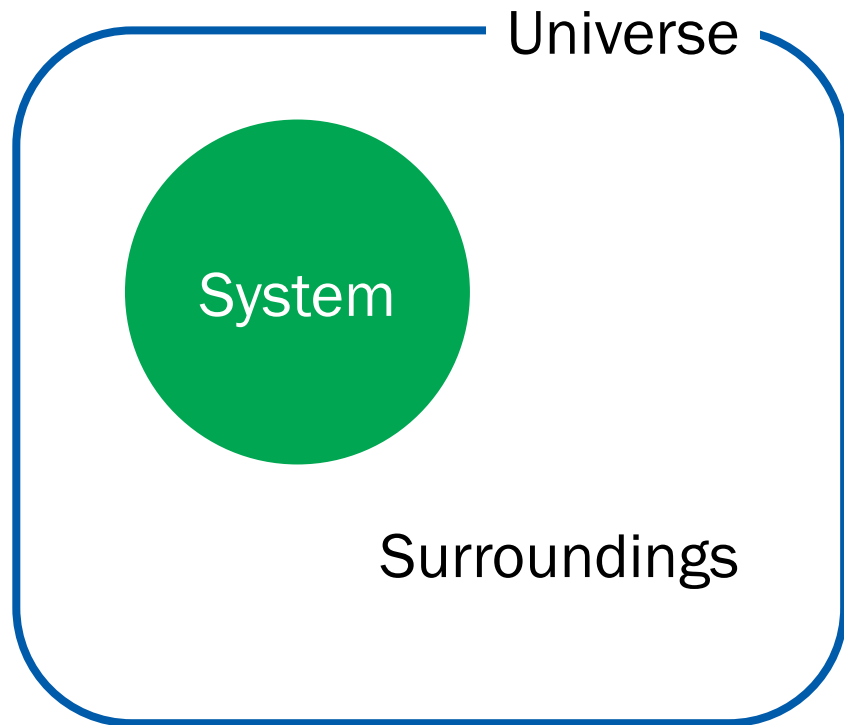


Third Example of a System

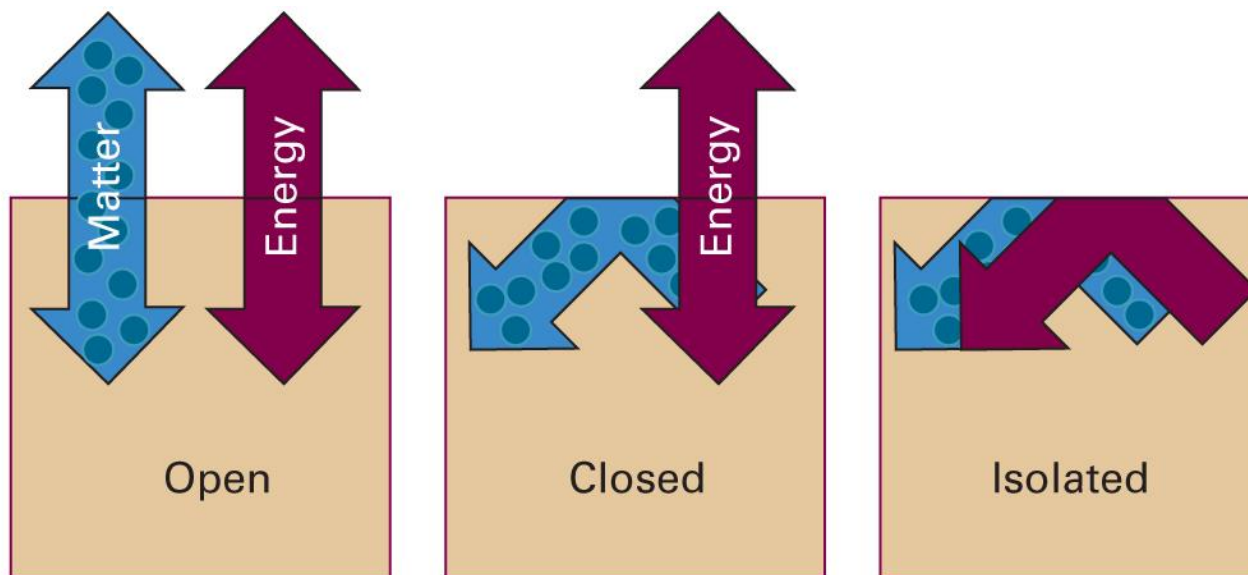


(Image: https://commons.wikimedia.org/wiki/File:Kitchen_design_at_a_store_in_NJ_refrigerator_and_cabinets_8.jpg)

And Even ...



Classification of Systems



System Properties

- A system can have different **physical properties**
- A **single value** will be assigned for each property
- Useful classification
 - **Extensive properties:** $f(A) + f(B) = f(A + B)$
 - **Intensive properties:** $f(A) + f(B) \neq f(A + B)$

Some Extensive Properties

- **Mass m :** measure of the quantity of matter present (unit: kilogram, kg)
- **Volume V :** measure of the quantity of space the sample occupies (unit: cubic meter, m³)
- **Amount of substance n :** measure of the number of specified entities (unit: mole, mol)

Relation between m and n

$$n \propto m$$

- Conversion: molar mass of a substance M (unit: kg mol^{-1})

$$n = \frac{m}{M}$$

Some Intensive Properties

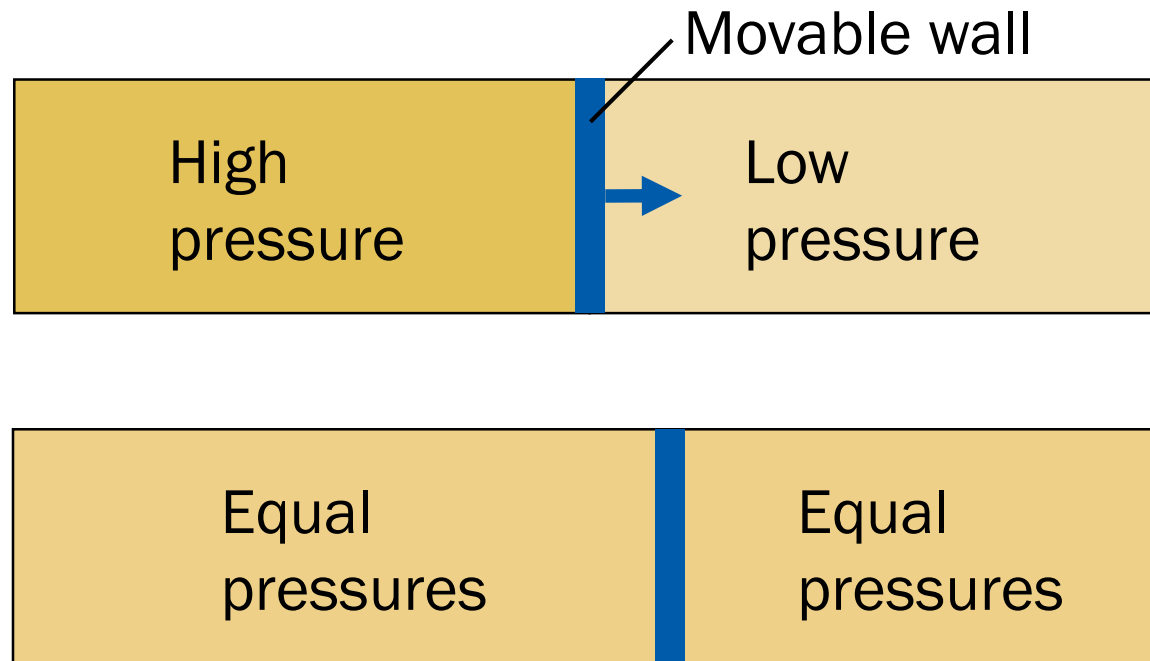
- **Pressure p :** force F / area A (unit: pascal, Pa)
 - Standard pressure $p^\ominus = 10^5 \text{ Pa} = 1 \text{ bar}$
 - Atmospheric pressure $101325 \text{ Pa} = 1 \text{ atm}$
- **Temperature T :** determines the direction of energy flow (unit: kelvin, K)
 - Celsius scale $T/\text{K} = \theta/^\circ\text{C} + 273.15$

Equilibrium

- An isolated system is **in equilibrium** when its properties remain **constant with time**
- **Equilibrium thermodynamics:** we will mostly focus on systems in equilibrium

Mechanical Equilibrium

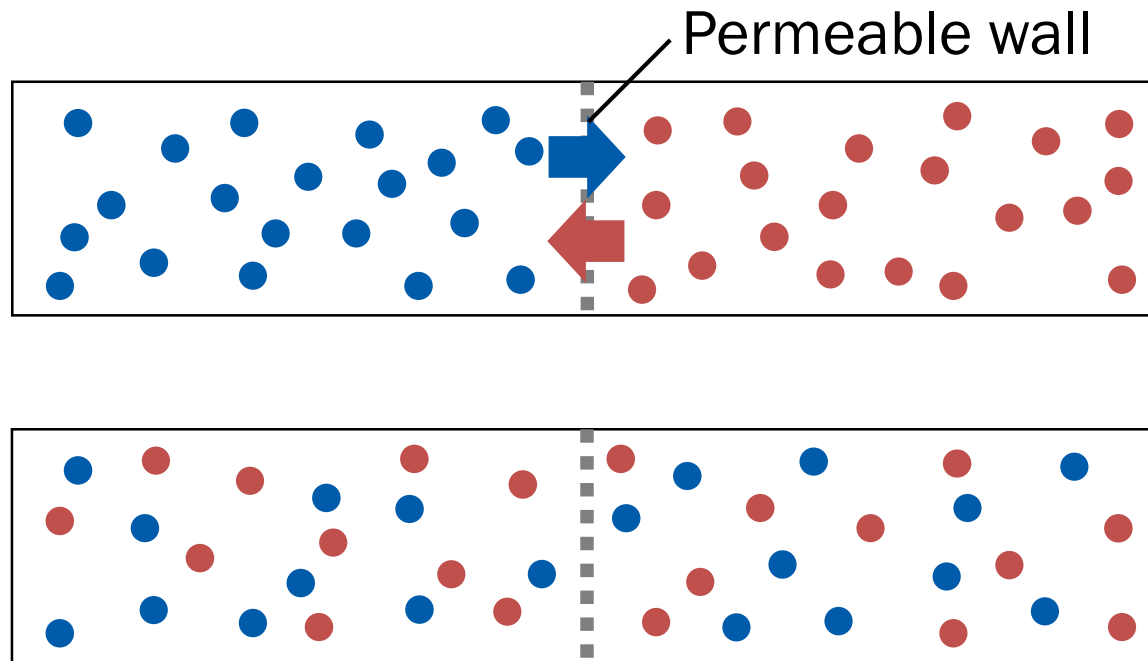
Constant p and V



Not in Atkins

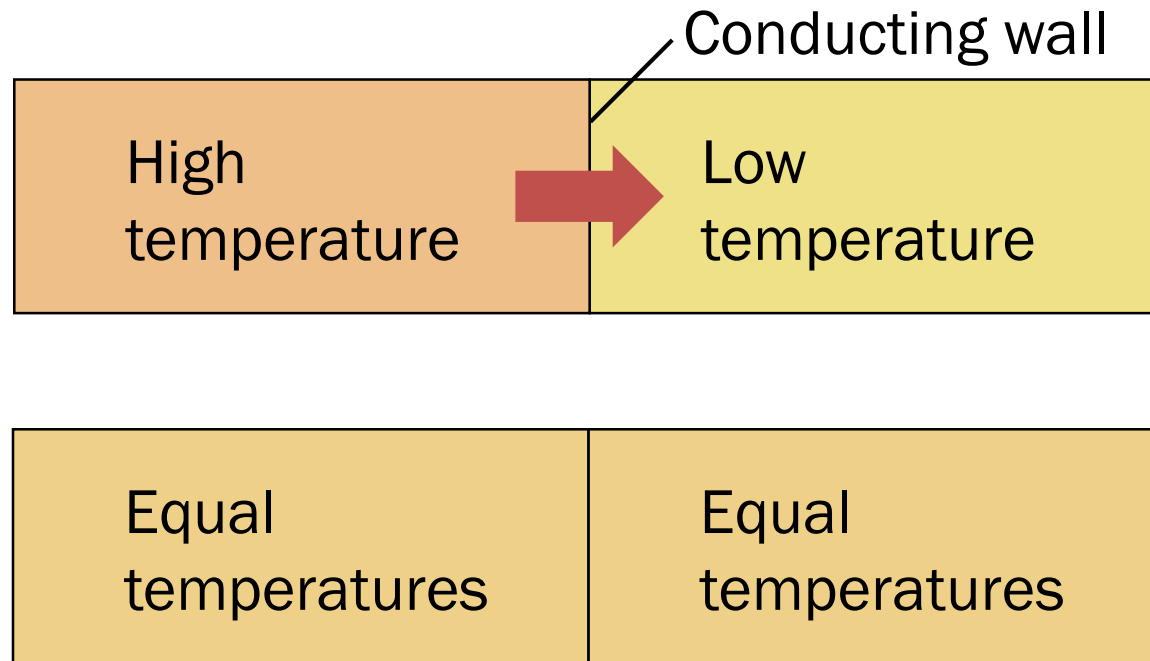
Diffusive Equilibrium

Constant n



Thermal Equilibrium

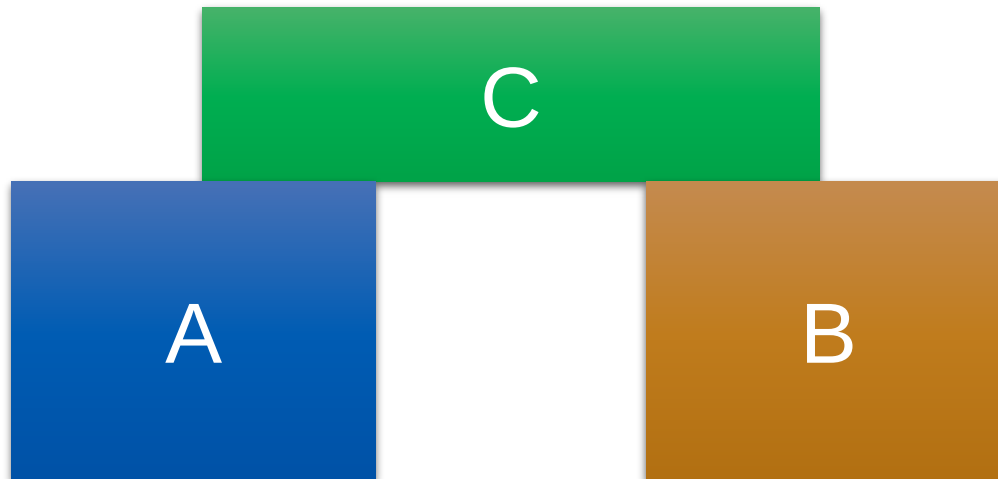
Constant T



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0th Law of Thermodynamics

“If systems **A** and **C** are in thermal equilibrium,
and systems **C** and **B** are in thermal equilibrium,
systems **A** and **B** are in thermal equilibrium.”



State of a System

- The **state** of a system is defined by its **properties**
- Two samples of **the same substance** that have **the same properties** are in **the same state**
- Pure substance: (n, V, p, T)

State Change (Process)

The system can change its state
from one to another

$$(n_1, V_1, p_1, T_1) \rightarrow (n_2, V_2, p_2, T_2)$$

Equation of State

- The state is defined by (n, V, p, T)
- **Equation of state:** relationship of the 4 variables

$$p = f(T, V, n)$$

- Depends on a system → We need a “model”

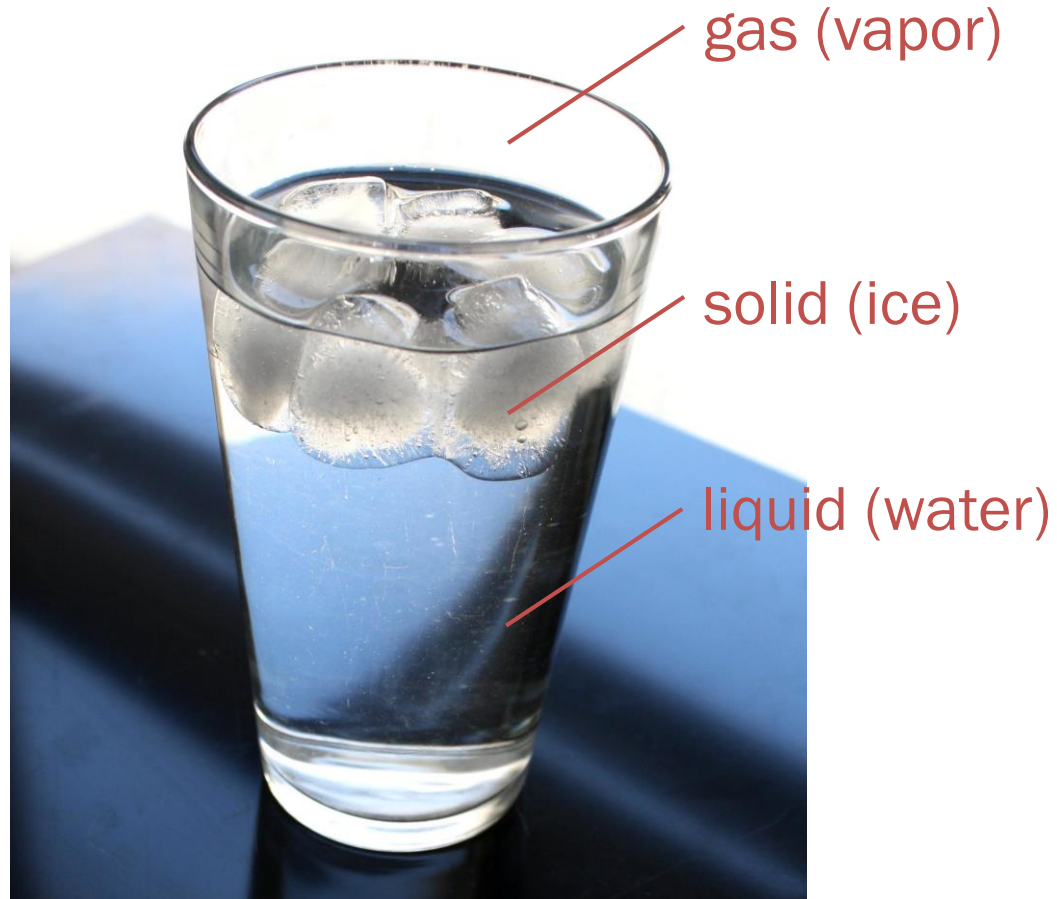
2. Gases

Chapter Overview

- Motivation: why gases?
- Equation of state: perfect gas
- Molecular kinetic theory
 - Maxwell-Boltzmann distribution
 - Collisions: molecule-molecule, molecule-wall
 - Kinetic energy of a gas
- Real gas

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Why Gases?

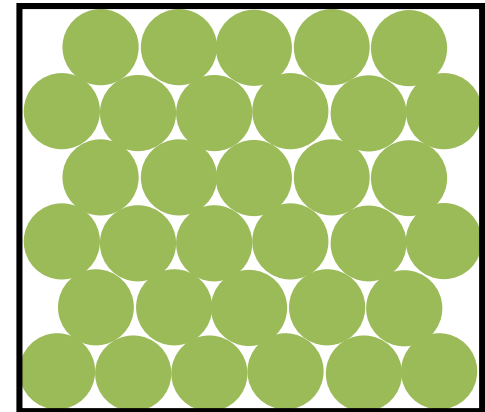
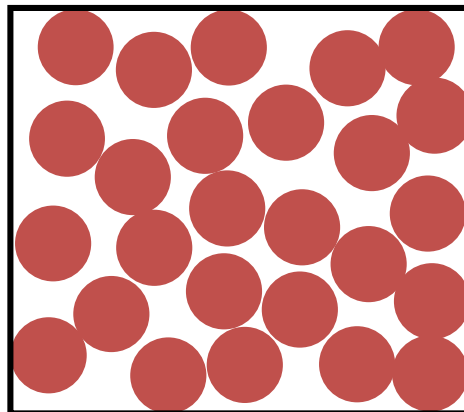
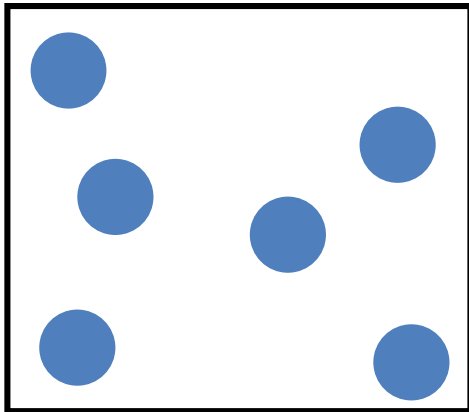


(Image: <https://pixnio.com/objects/glass/glass-ice-water>)

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Gas, Liquid, and Solid

condensed phase



Gas phase: kinetic energy $\gg$ potential energy

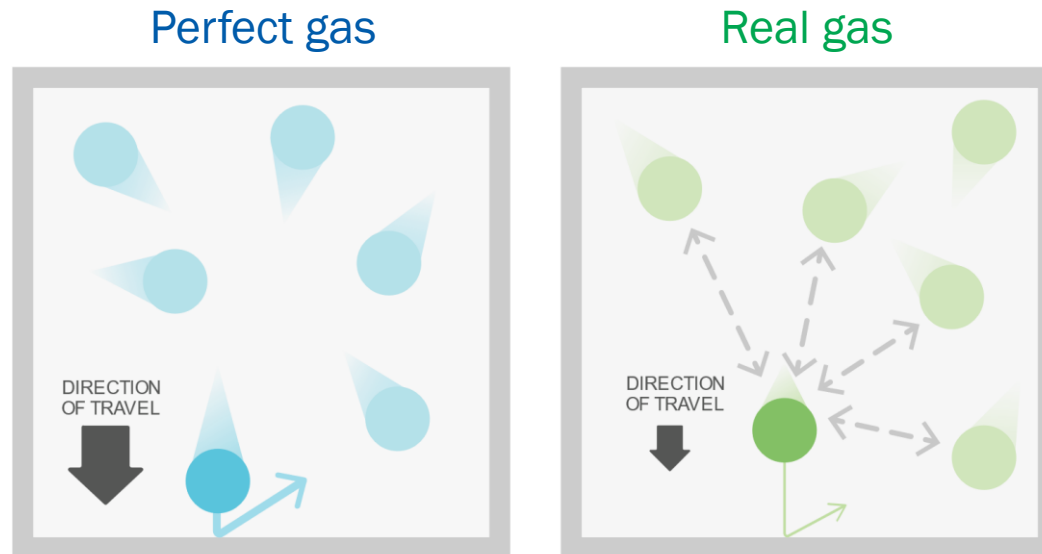
Condensed phase: kinetic energy $\approx$ potential energy

$$\frac{1}{2}mv^2$$

various forms

Perfect Gas

- Perfect gas (ideal gas): no interactions
- Real gas: finite interactions between molecules



(Image: <https://cdn.kastatic.org/ka-perseus-images/4d1bf2d46543cc34f4948085bb9196f5dfb57515.svg>)

Perfect Gas Law

- **Equation of state:** relationship of the 4 variables

$$p = f(T, V, n)$$

- Equation of state of a perfect gas: **perfect gas law**

$$p = nRT/V$$

↑
gas constant

Properties of Perfect Gas Law

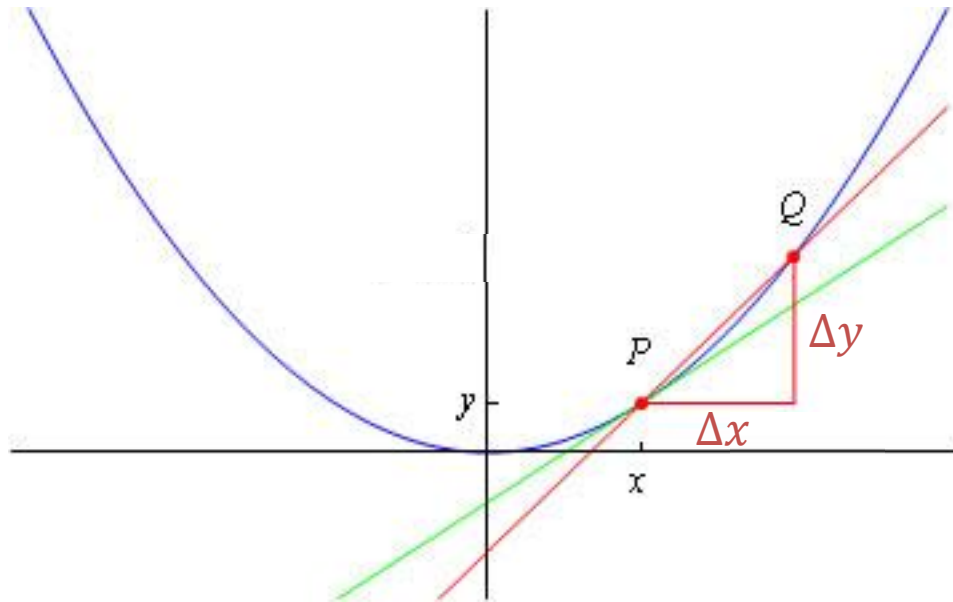
- Limiting law: in the limit of $p \rightarrow 0$, all real gases behave as a perfect gas
- STP (standard temperature and pressure)
 - $p = 1 \text{ atm}$, $T = 0^\circ \text{C} = 273.15 \text{ K}$
 - $V/n = 22.414 \text{ L mol}^{-1}$
- SATP (standard ambient temperature and pressure)
 - $p = 10^5 \text{ Pa}$, $T = 25^\circ \text{C} = 298.15 \text{ K}$
 - $V/n = 24.789 \text{ L mol}^{-1}$

Intermission: Review of Calculus



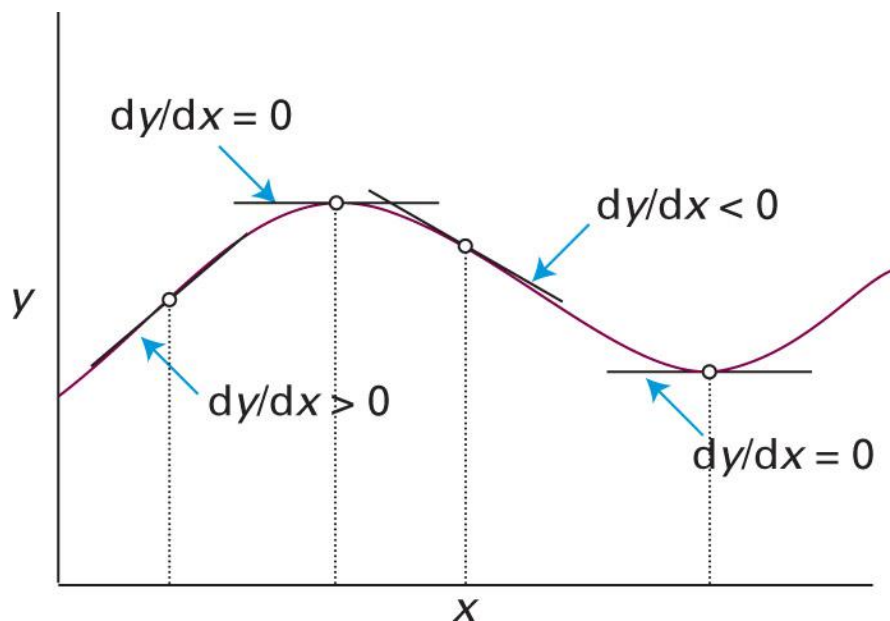
Differentiation

“The slope of a function”

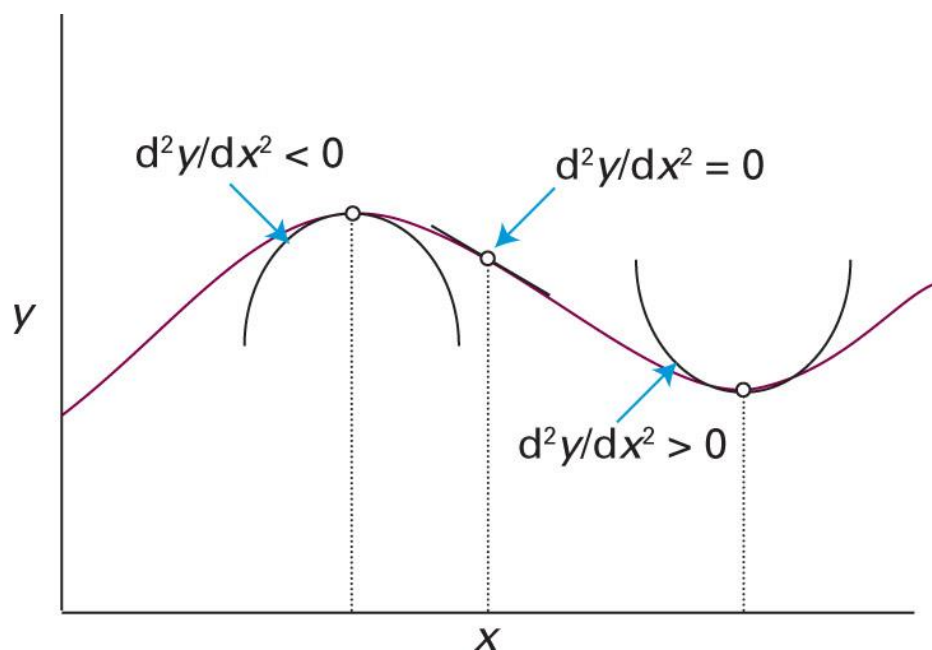


$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$$

First Derivative: Slope

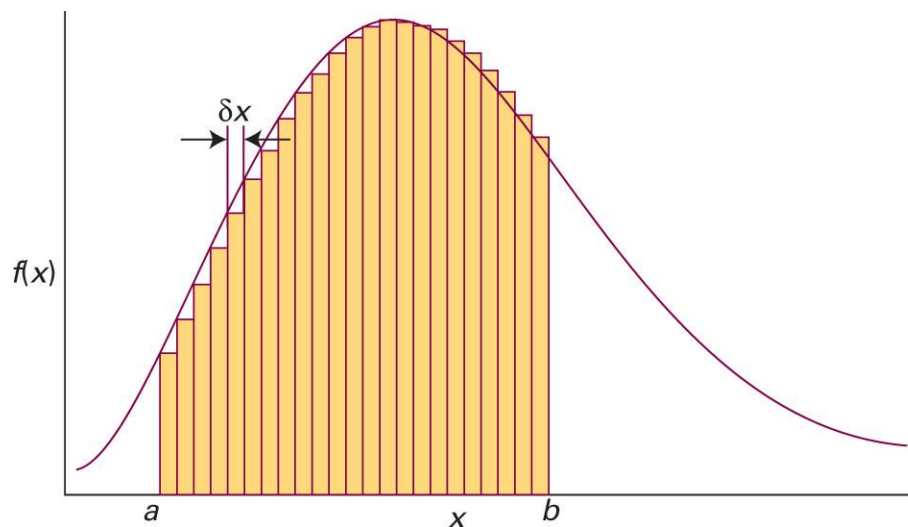


Second Derivative: Sharpness



Integration

“The areas under curves”



$$\lim_{\delta x \rightarrow 0} \sum_i f(x_i) \delta x = \int_a^b f(x) dx$$

How can we explain
the behavior of perfect gas?

Kinetic Molecular Theory

Total energy = Σ (kinetic energies of molecules)

Speed distribution

Collision frequency

Pressure

Internal Energy

⋮

Fundamental Assumptions

1. Molecules with mass m in ceaseless random motion
2. Molecules with negligible size
3. Elastic collisions: the total kinetic energy is conserved

Described by classical mechanics

Position and Velocity

Each particle with mass m has its own

- Position $\vec{r} = (x, y, z)$
- Velocity $\vec{v} = (v_x, v_y, v_z)$

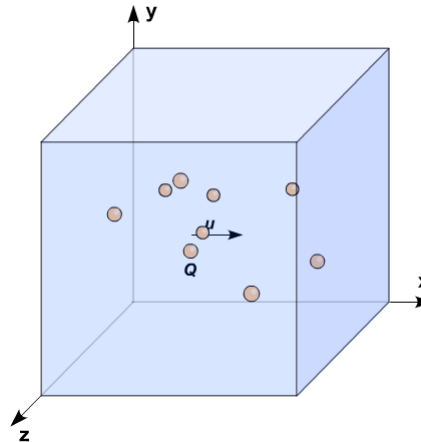
$$\vec{v} = \frac{d\vec{r}}{dt} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$$

With the velocity, we can define:

- Speed $v = |\vec{v}| = (v_x^2 + v_y^2 + v_z^2)^{1/2}$
- Kinetic energy $E_k = mv^2/2 = m(v_x^2 + v_y^2 + v_z^2)/2$

Molecular Velocities

Each molecule has a different velocity



How can we determine the information about velocity?

Probability: Example

Assume that we roll a fair dice. Probability is denoted as $p(k)$.

$$p(1) = p(2) = p(3) = p(4) = p(5) = p(6) = 1/6$$

- **Normalization:** $p(1) + p(2) + p(3) + p(4) + p(5) + p(6) = 1$

$$\sum_{k=1}^6 p(k) = 1$$

- **Average:** $\langle k \rangle = (1 + 2 + 3 + 4 + 5 + 6)/6 = 3.5$

$$\langle k \rangle = 1 \cdot p(1) + 2 \cdot p(2) + 3 \cdot p(3) + 4 \cdot p(4) + 5 \cdot p(5) + 6 \cdot p(6)$$

$$= \sum_{k=1}^6 k \cdot p(k)$$

Probability: Example

Now, we play a dice game with the following rule for a gain:

$$f(1) = \cancel{\text{¥}}10,000, f(2) = \cancel{\text{¥}}5,000, f(3) = f(4) = f(5) = f(6) = 0$$

- Expectation for the gain:

$$\langle f \rangle = \cancel{\text{¥}}10,000 \times (1/6) + \cancel{\text{¥}}5,000 \times (1/6) + 0 \times (4/6) = \cancel{\text{¥}}2,500$$

$$\begin{aligned}\langle f \rangle &= f(1)p(1) + f(2)p(2) + f(3)p(3) \\ &\quad + f(4)p(4) + f(5)p(5) + f(6)p(6) \\ &= \sum_{k=1}^6 f(k)p(k)\end{aligned}$$

Summary

	Discrete
Variable	$k = 1, 2, \dots, N$
Probability	$p(k)$
Normalization	$\sum_{k=1}^N p(k) = 1$
Average	$\langle k \rangle = \sum_{k=1}^N kp(k)$
Expectation	$\langle f \rangle = \sum_{k=1}^N f(k)p(k)$

From Discrete to Continuous

	Discrete	Continuous
Variable	$k = 1, 2, \dots, N$	x : real in $[x_1, x_2]$
Probability	$p(k)$	
Normalization	$\sum_{k=1}^N p(k) = 1$	
Average	$\langle k \rangle = \sum_{k=1}^N kp(k)$	
Expectation	$\langle f \rangle = \sum_{k=1}^N f(k)p(k)$	

Probability Distributions

	Discrete	Continuous
Variable	$k = 1, 2, \dots, N$	x : real in $[x_1, x_2]$
Probability	$p(k)$	$p(x)dx$ probability density ↘
Normalization	$\sum_{k=1}^N p(k) = 1$	$\int_{x_1}^{x_2} p(x) dx = 1$
Average	$\langle k \rangle = \sum_{k=1}^N kp(k)$	$\langle x \rangle = \int_{x_1}^{x_2} xp(x) dx$
Expectation	$\langle f \rangle = \sum_{k=1}^N f(k)p(k)$	$\langle f \rangle = \int_{x_1}^{x_2} f(x)p(x) dx$

Properties of Probabilities

- $\langle kf(x) \rangle = k\langle f(x) \rangle$

$$\langle kf(x) \rangle = \int_{x_1}^{x_2} [kf(x)]p(x) dx = k \int_{x_1}^{x_2} f(x)p(x) dx = k\langle f(x) \rangle$$

- $\langle f(x) + g(x) \rangle = \langle f(x) \rangle + \langle g(x) \rangle$

$$\begin{aligned}\langle f(x) + g(x) \rangle &= \int_{x_1}^{x_2} [f(x) + g(x)]p(x) dx \\ &= \int_{x_1}^{x_2} f(x)p(x) dx + \int_{x_1}^{x_2} g(x)p(x) dx = \langle f(x) \rangle + \langle g(x) \rangle\end{aligned}$$

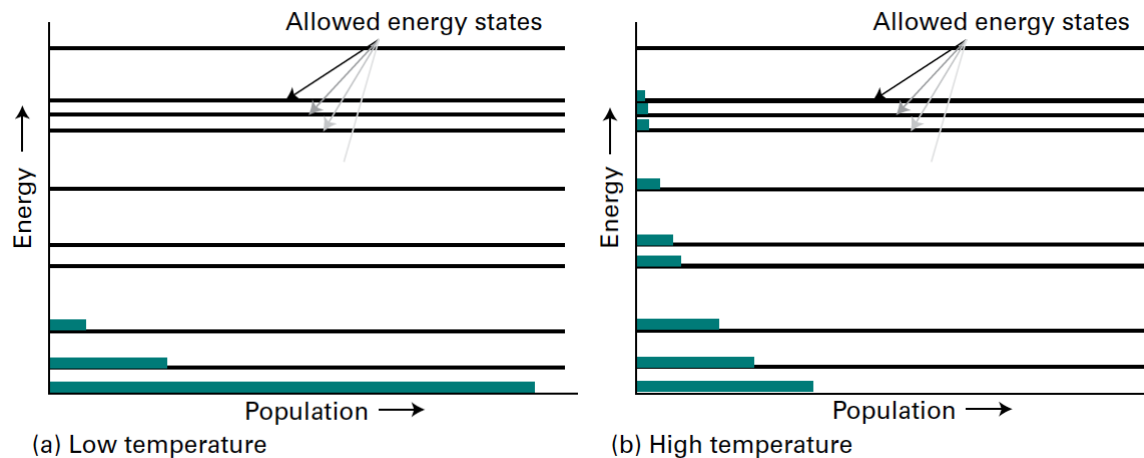
Boltzmann Distribution

Boltzmann constant



$$\text{Probability} \propto e^{-E/k_B T}$$

(at thermal equilibrium)



Boltzmann Distribution

- Probability density for energy E

$$f(E) \propto e^{-E/k_{\text{B}}T}$$

$$f(E) = K e^{-E/k_{\text{B}}T}$$

- $f(E)dE$: the probability that a molecule has an energy in the range $[E, E + dE]$

Probability Density for Velocity

$$f(E) = K e^{-E/k_B T}$$

Since the energy of a molecule = its kinetic energy,

$$E = \frac{1}{2} m v^2 = \frac{1}{2} m v_x^2 + \frac{1}{2} m v_y^2 + \frac{1}{2} m v_z^2$$

Probability density for velocity (v_x, v_y, v_z) :

$$f(v_x, v_y, v_z) = K e^{-m v_x^2 / 2 k_B T} e^{-m v_y^2 / 2 k_B T} e^{-m v_z^2 / 2 k_B T}$$

Separation of Variables

$$f(v_x, v_y, v_z) = K e^{-mv_x^2/2k_B T} e^{-mv_y^2/2k_B T} e^{-mv_z^2/2k_B T}$$

Let $f(v_x, v_y, v_z) = f(v_x)f(v_y)f(v_z)$, where

$$f(v_x) = K_x e^{-mv_x^2/2k_B T}$$

$$f(v_y) = K_y e^{-mv_y^2/2k_B T}$$

$$f(v_z) = K_z e^{-mv_z^2/2k_B T}$$

such that

$$K = K_x K_y K_z$$

Determination of K 's

$$f(v_x) = K_x e^{-mv_x^2/2k_B T}$$

Use the normalization condition.

Since the range of v_x is $(-\infty, \infty)$,

$$\int_{-\infty}^{\infty} f(v_x) dv_x = 1$$

$$\int_{-\infty}^{\infty} K_x e^{-mv_x^2/2k_B T} dv_x = 1$$

Determination of K 's

$$\int_{-\infty}^{\infty} f(v_x) dv_x = 1$$

$$\int_{-\infty}^{\infty} K_x e^{-mv_x^2/2k_B T} dv_x = 1$$

$$K_x \int_{-\infty}^{\infty} e^{-mv_x^2/2k_B T} dv_x = 1$$

$$K_x \left(\frac{2\pi k_B T}{m} \right)^{1/2} = 1$$

$$K_x = \left(\frac{m}{2\pi k_B T} \right)^{1/2}$$

Normalized Expression

$$K_x = \left(\frac{m}{2\pi k_B T} \right)^{1/2} = K_y = K_z$$

$$\begin{aligned} f(v_x, v_y, v_z) &= K_x K_y K_z e^{-mv_x^2/2k_B T} e^{-mv_y^2/2k_B T} e^{-mv_z^2/2k_B T} \\ &= \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-mv_x^2/2k_B T} e^{-mv_y^2/2k_B T} e^{-mv_z^2/2k_B T} \\ &= \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-m(v_x^2 + v_y^2 + v_z^2)/2k_B T} \end{aligned}$$